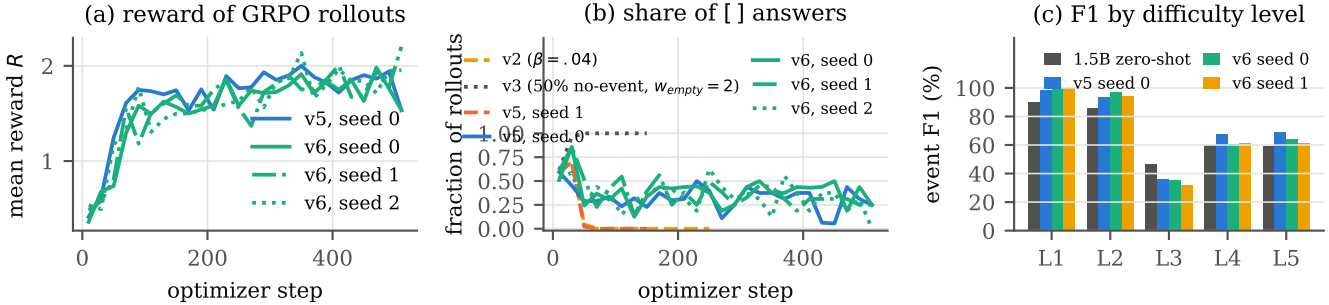
(a) reward of GRPO rollouts
mean reward R
optimizer step
v5, seed 0
v6, seed 0
v6, seed 1
v6, seed 2
(b) share of [ ] answers
fraction of rollouts
optimizer step
v2 ($\beta = .04$)
v3 (50% no-event, $w_{empty} = 2$)
v5, seed 1
v5, seed 0
v6, seed 0
v6, seed 1
v6, seed 2
(c) F1 by difficulty level
event F1 (%)
1.5B zero-shot
v5 seed 0
v6 seed 0
v6 seed 1
L1
L2
L3
L4
L5